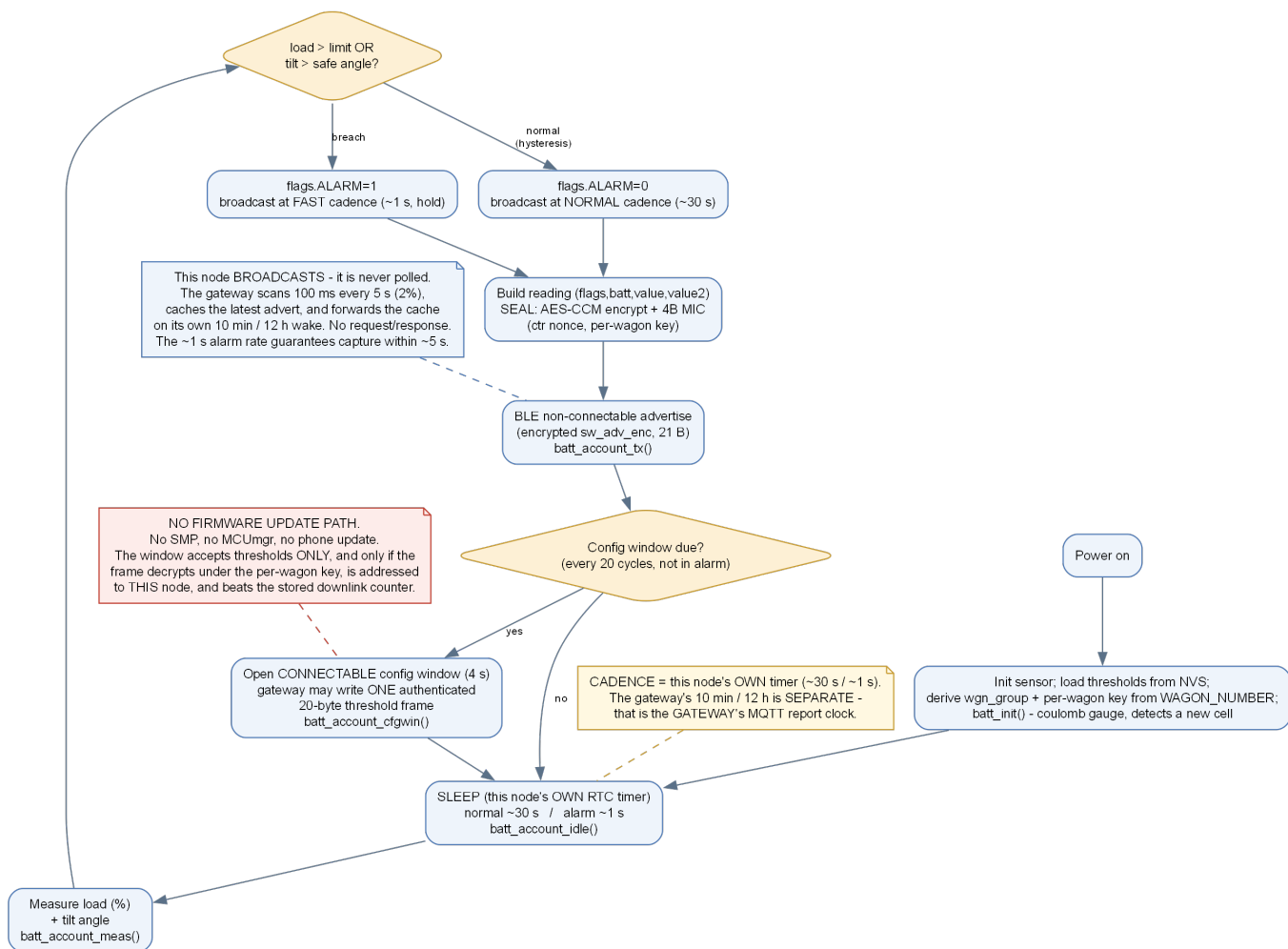


Sub-node — Load & Tilt Detection

4 per wagon · load status + tilt angle



NOT IMPLEMENTED IN FIRMWARE. This sheet describes the *target* design. In the current build this node type has no sensor driver: `sensor.c` is a placeholder that returns a fixed 25.0 °C and a zero secondary value, and its overlay declares no I2C bus, so the BMA400 and STS4x fitted to the board are not read. Everything else on this sheet - BLE, AES-CCM, the battery gauge, the config window, the cadence logic - IS implemented and working. Do not field this node type until the sensor driver lands.

1. Role

A battery-powered, **connectionless BLE broadcaster**. It measures its sensor, **encrypts** the reading into a BLE advertisement (AES-CCM, with the wagon-group id for isolation), broadcasts a short burst, and sleeps. On a threshold breach it switches to fast "alarm" advertising so the gateway's receive window catches it within ~5 s. It forms a connection only during a periodic **config window**, and only to receive thresholds.

Timing — no polling: this node advertises on its **own** timer — about every **30 s** normally, **~1 s** while in alarm. The gateway's **10 min / 12 h** cadence is a **separate** clock. This node is **never polled**: it broadcasts blind, the gateway's low-power scan (**100 ms every 5 s, 2% duty**) caches the latest advert, and on the gateway's own schedule it forwards that cache upward.

2. Algorithm

1. Init: load `node_id`, `wgn_group` and thresholds from NVS; start the coulomb gauge (which auto-zeroes if it detects a replacement cell).
2. Wake on the node's own RTC timer; measure load (%) + tilt angle.
3. Evaluate against the alarm threshold **with hysteresis** (on angle $\geq$ set, off $<$ set-hyst).

4. **Seal** the reading with AES-CCM under the per-wagon key, using a fresh monotonic counter `ctr` as the nonce (persisted in NVS so it never repeats). Nonce byte 7 carries a direction tag so an uplink nonce can never collide with a downlink one.
5. Advertise the encrypted burst (non-connectable); **slow when normal (~30 s), fast while in alarm (~1 s)**.
6. Every 20 normal cycles (never during an alarm) open a 4 s connectable **config window** so the gateway can write new thresholds.
No firmware can be loaded through it.

3. Advertisement payload (encrypted `sw_adv_enc`, 21 B)

Cleartext header (company, ver, wgn_group, type, id, ctr) + **AES-CCM ciphertext** of the fields below + 4-byte MIC. The header is authenticated; the reading itself is encrypted.

Field	Meaning for this node
wgn_group	This wagon's isolation group, from the wagon number (gateway drops other groups; cleartext)
node_type / node_id	sensor type + unique id within the wagon (cleartext header)
ctr	monotonic nonce counter, persisted in NVS (cleartext; makes each advert unique)
flags / batt (encrypted)	ALARM/LOWBATT flags, battery % from the coulomb gauge
value	load (%)
value2	tilt ×10 (deg)

4. Alarm / event mapping

Condition	Protocol code	Notes
Load over configured limit	OVERLOAD	sev warn/crit
Tilted / asymmetric load	TILT	safety
Loaded / unloaded at yard	LOAD_CHANGE (event)	state change

5. Battery gauge (coulomb counting)

The cell is a **non-rechargeable 13 Ah Li-SOCl₂; ER34615**. Its discharge curve is flat, so a voltage-to-percent map reads ~100 % for years and then collapses. State of charge is therefore obtained by **integrating what the firmware spends**: every sleep, measurement, advertising burst and config window adds (current × time) in nAh to an accumulator persisted in NVS. Self-discharge is carried as a constant current so elapsed time is never free.

Item	Value
Usable capacity	BATT_USABLE_MAH = 10000 (derated from the 13 Ah nameplate for cold and load)
Total current ceiling	13 Ah over a 6-year life = 247 µA average
Predicted average	~134 µA (sleep 5 + self-discharge 15 + measure 1 + advertising 60 + config window 53)
Independent end-of-life check	BATT_EOL_MV = 3000 mV, so counter drift cannot hide a dying cell

Calibration outstanding: every `BATT_I*_UA` constant except self-discharge is an estimate marked `MEASURE:` in the source. They must be measured with a Nordic PPK2 in series with the cell over one full wake cycle before the projected life means anything.

6. What is missing

- Tilt can be derived directly from the fitted BMA400 (gravity vector → angle), but no such code exists on this node.
- Load sensing has no defined transducer on this board - no schematic supplied.
- Devicetree: no I2C bus, no sensor rail switch, no BMA400 interrupt line.

Ref: RDSO WD-35-MISC-2024 · SmartWagon Telemetry Protocol Rev.1 (pv=1) · controller nRF54L15 (QFN52). Generated from the firmware sources by `DOCS/gen_pdfs.py` - regenerate after changing the code.